

Ahmet Levent

ZENGİN



FOCUS

Mechanical Engineer specializing in CFD, wind energy technologies, and data-driven design.

CONTACT DETAILS

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cozygradient.github.io

Ankara, TR

LANGUAGES

English (Fluent, IELTS 8.0)

Turkish (Native)

German (A2)

SKILLS

- **Programming:** Python, C++, Rust
- **Data & ML:** pandas, NumPy, scikit-learn
- **CAD:** SolidWorks, AutoCAD, MATLAB
- **Simulation:** ANSYS Fluent, OpenFOAM

PROJECT MANAGEMENT

CAPM Training

(Certified Associate in Project Management)

In progress

CERTIFICATIONS

UAV-1 Commercial Drone Pilot

SHGM, issued May 2026

PROFILE

A mechanical engineer who enjoys solving practical problems and turning ideas into working solutions. Comfortable simulating and testing designs before they're built, and skilled at translating technical findings into clear, actionable recommendations for both engineers and non-engineers. Works well within teams, communicates clearly across departments, and adapts quickly when priorities shift. Currently building further strengths in data analysis and project management to support well-organized, collaborative delivery from concept to completion.

EDUCATION

B.Sc. in Mechanical Engineering

Başkent University

2021–2026

- EUR-ACE® Accredited Program
- Relevant Coursework: Aerodynamics, Computational Fluid Dynamics

EXPERIENCE

Language Mentor (Volunteer) at Balakun

2026 – Present

- Facilitated weekly English conversation sessions for Ukrainian learners, adapting lesson plans to diverse proficiency levels and learning goals.
- Integrated into a global volunteer mentor network to support international educational outreach.

Office Internship at Kaleliler Fasteners

Jan – Feb 2026

- Streamlined engineering operations by managing technical documentation, updating product specifications, and generating production reports.
- Ensured compliance with international industrial fastener standards during product specification reviews.

Manufacturing Internship at UNF Makina

Aug – Sept 2024

- Executed quality control inspections and analyzed manufacturing workflows to identify process bottlenecks and improve production efficiency.
- Observed and documented industrial manufacturing strategies and process flow optimizations.

Team Leader, Energy Society at Başkent University

2023–2025

- Spearheaded the 3D design and CFD simulation of steam turbine components to optimize aerodynamic efficiency.
- Directed a cross-functional student team, managing project timelines, resource allocation, and technical deliverables.

PUBLICATIONS

A Parametric Investigation of the Unloading Behaviour of Seats during Rear Impact

Oct 2026

- S. Himmetoglu, A. L. Zengin et al., *Transport Means 2026*, Kaunas, Lithuania.

The Effect of Flow Separation Angles on Compressor Pressure in Axial Compressor Blades

May 2025

- A. Koçer, E. Öztürk, A. L. Zengin, *9th International Students Science Congress*, Manisa, Turkey.

PORTFOLIO

🌐 cozygradient.github.io
GitHub projects & CFD visuals

KEY COURSEWORK

- Aerodynamics
- Computational Fluid Dynamics
- Thermodynamics
- Strength of Materials

INTERESTS

- Fluid dynamics visualisation
- UAV photography & flight planning
- Open-source tooling
- Industrial Design
- Film Making

AVAILABILITY

Open to relocation
Immediate start

PROJECTS

BERTopic Topic Modelling of Group Chats

Aug 2026

- Applied transformer-based topic modelling (BERTopic) to a group-chat corpus to surface recurring themes and conversation dynamics.
- Built an end-to-end Jupyter Notebook pipeline covering text preprocessing, embedding, and topic visualisation.

CozyGradient Portfolio Website

Jul 2026

- Designed and built a personal portfolio site from scratch to showcase projects, CFD visuals, and open-source work (cozygradient.github.io).

Small-Scale Open-Source Wind Tunnel

Oct 2025 – Jun 2026

- Designed and built a small-scale wind tunnel for low-speed aerodynamic testing and flow visualisation.
- Documented the full workflow from CAD modelling through fabrication under an open-source licence.

Factory Floor Plan Optimisation & 3D Rendering

Feb 2026

- Analysed production facility layouts and produced 3D renderings to evaluate workflow, space utilisation, and process flow.

Analysis of Axial Compressors with CFD

May 2024

- Simulated flow through axial compressor blade geometries to assess pressure rise, flow separation, and blade aerodynamic performance.

ADDITIONAL EDUCATION

Data Analysis & Machine Learning with Python

Independent Coursework

2026 – Present

- Building proficiency in pandas, NumPy, and scikit-learn to construct predictive models for engineering datasets.

An Introduction to Computational Fluid Dynamics

Udemy (Instructor: Sebastian Thomas)

2025

- Applied CFD principles to solve real-world fluid dynamics and heat transfer problems using industry-standard software.